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Bihar Engineering University, Patna

End Semester Examination - 2022

Subject: Hydrology & Water resource engineering

Code: 101505

Time: 03 Hours

Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option/answer of the following (Any seven question only): **[14]**

(a) In the hydrological cycle the average residence time of water in the global
- (i) Atmospheric moisture is larger than that in the global rivers
- (ii) Oceans is smaller than that of the global ground water
- (iii) Rivers is larger than that of the global ground water
- (iv) Oceans is larger than that of the global ground water

(b) The double mass curve technique is adopted to
- (i) check the consistency of rain gauge records
- (ii) find the average rainfall over a number of years
- (iii) find the number of rain gauges required
- (iv) estimate the missing rainfall data.

(c) An Isohyet is a line joining points having
- (i) equal evaporation value
- (ii) equal barometric pressure
- (iii) equal height above the MSL
- (iv) equal rainfall depth in a given duration

(d) Which instrument is used for measurement of Evapotranspiration?
- (i) Lysimeter
- (ii) Atmometer
- (iii) Phytometer
- (iv) Hygrometer

(e) A hydrograph is a plot of
- (i) Rainfall intensity against time
- (ii) Stream discharge against time.
- (iii) Cumulative rainfall against time
- (iv) Cumulative runoff against time

(f) The total rainfall in a catchment of area 1200 km^2 during a 6-h storm is 16 cm while the surface runoff due to the storm is 1.2×10^3 m^3. The φ-index is
- (i) 0.1 cm/h
- (ii) 1.0 cm/h
- (iii) 0.2 cm/h
- (iv) cannot be estimated

(g) The discharge per unit drawdown at the well is known as
- (i) Specific Capacity
- (ii) Specific storage
- (iii) Specific retention
- (iv) Specific yield

(h) The relationship between the duty D in hectare/cumec, the water depth Δ in centimetre, and base period B in days, is given by:
- (i) $D = [8.64\times B] / [\Delta]$
- (ii) $D = [864\times B] / [\Delta]$
- (iii) $D = [8.64\times \Delta] / [B]$
- (iv) $D = [864\times \Delta] / [B]$

(i) The Garret's diagrams are based on:
- (i) Lacey's theory
- (ii) Khosla's theory
- (iii) Bligh's theory
- (iv) Kennedy's theory

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(j) During the maintenance of an earthen dam, the apparent seepage through the foundation of the dam is best taken care of, by providing:

(i) a chimney drain
(ii) a rock toe
(iii) a drain trench along the downstream toe.
(iv) an upstream impervious cutoff

Q.2 Solve both questions: **[14]**

(a) Differentiate between: (i) confined and unconfined aquifer and (ii) aquiclude and aquitard.

(b) A 20 cm diameter well fully penetrates a confined aquifer of thickness 25 m when the well is pumped at a 200 litres/minute. The steady state drawdown in two observation wells located at 10 m and 100 m distance from pumping well are found to be 3.5 m and 0.05 m, respectively. Calculate the permeability and transmissivity of the aquifer.

Q.3 Solve both questions: **[14]**

(a) Explain evapotranspiration and factors affecting evapotranspiration.

(b) A storm with a 15 cm precipitation produces a direct runoff of 8.7 cm. The time distribution of storm is as follows:

Time from start (in hrs)	1	2	3	4	5	6	7	8
Rainfall in each hr (cm)	0.6	1.35	2.25	3.45	2.7	2.4	1.5	0.75

Estimate the value of φ-index for the storm.

Q.4 Solve both questions: **[14]**

(a) Describe the salient characteristics of precipitation in India.

(b) The normal annual precipitation of five raingauge stations P, Q, R, S and T are respectively 125 cm, 102 cm, 76 cm, 113 cm, and 137 cm. During a particular storm, the precipitation recorded by stations P, Q, R and S are 13.2 cm, 9.2 cm, 6.8 cm and 10.2 cm respectively. The instrument at station T was inoperative during that storm. Estimate the rainfall at station T during that storm.

Q.5 Solve both questions: **[14]**

(a) Discuss in brief various methods of surface irrigation.

(b) A sandy loam soil holds water at 140 mm/m depth between field capacity and permanent wilting point. The root depth of the crop is 30 cm and the allowable depletion of water is 35%. The daily water use by the crop is 5 mm/day. The area to be irrigated is 60 hectare and water can be delivered at 28 litre per second. The surface irrigation application efficiency is 40%. There are no rainfall and ground water contribution. Determine i) allowable depletion depth between irrigations; ii) frequency of irrigation; iii) net application depth of water and iv) volume of water required.

Q.6 Solve all questions: **[14]**

(a) Define Paleo Irrigation and Kor Watering.

(b) Define Sprinkle Irrigation and Furrow Irrigation.

(c) A water course has a culturable command area of 5000 hectare out of which intensities of perennial sugarcane and rice crop are 25% and 50% respectively. The Duty of these crops at the head of water course are 1000 hectare/cumecs and 1500 hectare/cumecs respectively. Find the discharge required if the peak demand is 125% of the average requirement.

Q.7 Solve both questions: **[14]**

(a) What is meant by canal lining and also discuss its advantages?

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(b) A watershed has five non-recording rain gauges, installed in its area. The amount of rainfall recorded for one of the years is given below:

Rain Gauge Stations	P	Q	R	S	T
Annual Rainfall in cm	100	120	190	125	95

Find the required optimum number of non-recording and recording rain gauges for this watershed. Assume an error of 10% in the estimation of mean rainfall.

Q.8 Solve both questions: **[14]**

(a) Explain the forces acting on the gravity dam.

(b) Explain the causes and effects of waterlogging.

Q.9 Solve both questions: **[14]**

(a) Discuss in detail the application of unit hydrograph.

(b) Following are the ordinates of a storm hydrograph of a river draining a catchment area of 423 Km^2 due to a 6-h isolated storm. Derive the ordinates of a 6-h unit hydrograph for the catchment.

Time from start (h)	0	6	12	18	24	30	36	42	48
Discharge (m^3/s)	10	30	87.5	115.5	102.5	85	71	59	47.5
Time from start (h)	54	60	66	72	78	84	90	96	102
Discharge (m^3/s)	39	31.5	26	21.5	17.5	15	12.5	12	12

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